

A Distribution of Possibilities

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Calculate the paths for Next token distributions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Next token distributions. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Analyse a new table independently

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum. Which first token is greedy, which end path is largest, and does either verify a real status?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A toy next-token distribution is sun=.5, rain=.3, snow=.2. Over ten samples, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

6. Solve the given case

A toy next-token distribution is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A toy next-token distribution assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 2 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Explain that Study Buddy responses come from possibilities, so important claims need checks.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A toy next-token distribution is sun=.5, rain=.3, snow=.2. Over ten samples, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Validate the table

Check 0.5,0.3,0.2 as probabilities. Interpret their sum and the complement of ready.

My reasoning and evidence

13. W2 Calculate the paths

Compute ready,end; busy,end; unknown,end and their combined probability.

My reasoning and evidence

14. W3 Test a greedy claim

Use $A=0.6, B=0.4$ and end conditional probabilities 0.51 and 1. Compare greedy A,end with B,end.

My reasoning and evidence

15. W4 Analyse a new table independently

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum. Which first token is greedy, which end path is largest, and does either verify a real status?

My reasoning and evidence

16. W5 Normalise with a stated rule

Weights are ready 6,busy 3,unknown 1. Convert them into probabilities. What if a weight is -1?

My reasoning and evidence

17. Embeddings and position — W1 Look up vectors

For red blue name the IDs, embedding rows and shape before adding positions.

My reasoning and evidence

18. Embeddings and position — W2 Add positions

Compute representations for red blue and blue red under the stated position table.

My reasoning and evidence

19. Embeddings and position — W3 Audit the sum

Sum the rows for both sequences. Does the sum establish that the original text had the same order?

My reasoning and evidence

20. Embeddings and position — W4 Trace new vectors independently

Embeddings $A=[1,2]$, $B=[3,0]$, position 0= $[0,1]$, position 1= $[1,0]$. Compute represented rows and sums for A B and B A. Explain what the equal sums mean.

My reasoning and evidence

21. Embeddings and position — W5 Handle a shape failure

An embedding has width two but its position vector has width three. The input also contains a third position absent from the table. Specify honest handling.

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: A Distribution of Possibilities

A fictional two-stage token tree illustrates next-token prediction.

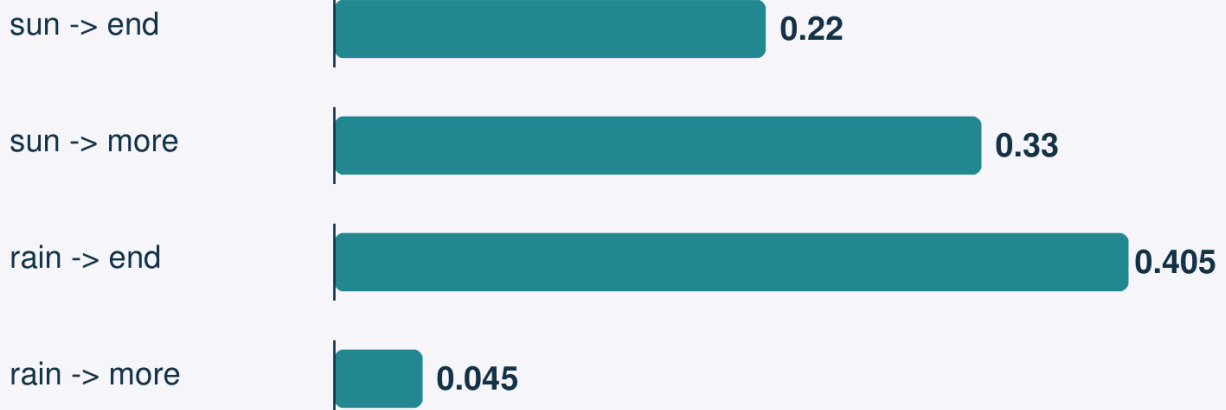
First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

AI ACADEMY / AI-3/2

Joint probabilities of the four toy branches

probability



Bar length shows the supplied value; the marked baseline is zero.

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end: $0.55 \times 0.4 = 0.22$; sun -> more: $0.55 \times 0.6 = 0.33$.



3. Rain branches

rain -> end: $0.45 \times 0.9 = 0.405$; rain -> more: $0.45 \times 0.1 = 0.045$.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Why is 0.625 not a broken probability sum?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

24. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional two-stage token tree illustrates next-token prediction.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? rain→end is the more probable of the two complete end paths.

1. Rain branches
2. Compare end paths
3. Sun branches
4. First distribution

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare end paths rain→end is the more probable of the two complete end paths. Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Practice 2: explain the deciding evidence

Changed-case practice: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. sun→end becomes 0.44, beating unchanged rain→end 0.405.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: sun→end becomes 0.44, beating unchanged rain→end 0.405. Changing one conditional distribution changes its path products while the first-token probabilities remain fixed.

Repair a missing building block

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
